

Final Project for Econometrics II – Replication work

Causal impact of unionisation on production quality

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Introduction

This paper replicates Kini et al (2022), investigating the causal impact of labour unions on production quality – measured by product recalls. I adopt a two-part Difference-in-differences (DiD) model with fixed-effects predicting the intensive and extensive margin effects of unionisation on recalls. I probe the validity of the parallel-trends assumption graphically.

Then I conduct an RDD which measures the difference in product recalls around the unionisation threshold of 50% votes using the two-part model.

Finally, for robustness, I test for reverse causality using lead and lag values of the unionisation variable in the DiD setting. In the RDD setting, I verify whether there is any discontinuity in the outcome variables associated with firm characteristics at the unionisation cutoff.

Causal challenges

A pooled OLS regression of product recalls on unionisation could suffer endogeneity through omitted variable bias, reverse causality and measurement errors. In this context, I expect omitted variable bias might be present in either direction. For example, firms with poor management quality might have higher recalls and might also lead workers to unionise to protect their interests. This would lead to a positive bias. However, higher expenditure on R&D might inspire workers to unionise against expected layoffs (workers being replaced by machines) but improve production quality and hence reduce recalls. This would bias estimates negatively.

While unionisation might cause product recalls, workers might also unionise to protect themselves against wage cuts after product recalls are observed. This would lead to a reverse causal relation in which product recalls in time (t) lead to unionisation in subsequent time periods ($t + n$). These estimates would be biased (in either direction) and inconsistent.

Measurement error has the potential to create attenuation bias in case of classical errors, or bias in any direction otherwise, but I cannot manage it.

Data and Measurement

I use a subset of variables from the data in Kini et al., which was collected from the Food and Drug Administration (FDA), Consumer Product Safety Commission (CPSC), and National Highway Traffic Safety Administration (NHTSA) from January 2003 to December 2013.

The outcome variable of interest is the number of cumulative product recalls in years $t + 1$, $t + 2$ and $t + 3$. I use these variables to construct a) the **probability** of any recalls occurring for the extensive margin response; and b) the **frequency** of positive recalls for the intensive margin response (reason for specification discussed in next section). Since the dependent variable is leading forward in time, all the independent variables are automatically lagged. These lagged independent variables ensure that unionisation has time to effect production quality. The sample is restricted to industries having at least one firm recall in the dataset.

Control variables for the panel include firm characteristics which generally explain production quality such as size, leverage, Herfindahl index, total factor productivity, vertical integration and R&D intensity. I omit reporting the table of summary statistics (which I made for my personal use) for this dataset since it has the same characteristics as the original paper. The union election data used for RDD is obtained by Kini et al, from Bloomberg BNA Labor Plus. It contains percentage of votes for unionisation, year, agency of recall, RTW (Right to Work) states and outcome variables like cost of goods to sales ratio, operating leverage, CSR ratings, ratio of capital, R&D and other expenses, to book value of assets.

Difference-in-differences

Empirical strategy

Since I have access to firm-year panel data, I intend to use a DiD strategy to estimate the impact of unionisation on product recalls over time. I would assume that in the absence of unionisation, treated and control firms would have followed parallel recall trajectories. However, there is no clear control and treatment group, since firms have different unionisation status over different years. Certain firms may be more likely to be unionised than others because of characteristics such as firm size, complexity of production, etc and might also have higher or lower product recalls due to these factors. Since the treatment is not randomised, the parallel trends assumption of the DiD is likely to be violated. Hence, I use the DiD with

industry-year fixed effects to account for unobserved heterogeneity in industry characteristics that change over time. Ideally, I would use firm and year fixed effects. However, Firm fixed effects absorb nearly all within-firm variation in the unionisation variable, causing Stata to drop it from the estimation due to collinearity.

While Kini et al. use a Poisson log-linear model, approximately 95% of dependent variable – product recalls – is zero and subsequent frequencies are severely right-skewed. Given this over-dispersion (the variance is much larger than the mean), it is useful to verify the result using alternate model specifications. A linear model would not be suitable since it assigns a single linear conditional mean to the number of recalls. A transformation of $\log(x+1)$, or inverse hyperbolic sin function would reduce the right skewedness but still not solve the problem of inflated zeros. Thus, I propose a two-part model: first considering the **extensive** margin – probability of experiencing at least one recall, and then considering the **intensive** margin – expected number of (non-zero, positive) recalls. This allows distinct models to explain the seemingly different data-generating processes of product recalls. In the extensive margin, collecting all the positive recalls under one bin reduces the variability of the model thus improving prediction accuracy. In the intensive margin, we measure the factors impacting the frequency of recalls, conditional on recalls being positive. Focusing on the positive recalls subsample enables the model to predict the factors correlated with the increasing frequency of recalls, and not just the jump from non-recalls to positive recalls. I acknowledge that the unbiasedness of these results is caveated due to the selection on positive frequency of recalls.

I estimate the DiD using a linear probability model (LPM) and ordinary least squares (OLS) with industry-year fixed effects. While logistic models are standard for binary outcomes, their fixed-effects implementation drops units with no within variation in the dependent variable, which is particularly problematic in this setting where product recalls are rare and highly persistent. As a result, a fixed-effects logit would substantially reduce the effective sample and limit identification to a smaller subset of firms.

The main independent variable of interest – the percentage of workers under union contracts – is zero for approximately 90% firms, with subsequently tapering frequency. Again,

I consider an intensive and extensive margin response characterisation: I use a binary variable (ifunion_len3) that is set to one for firms with any unionised workers assuming contract lengths are 3 years. Then, I consider the percentage of unionisation, including the zero

Extensive Margin:

$$\text{Recall dummy}_{i,j,t+n} = \beta_0 + \beta_1 \cdot \text{Unionised}_{i,t} + X_{i,j,t} + \text{Industry}_j \times \text{Year}_t + \varepsilon_{i,j,t}$$

Intensive Margin:

$$\text{Number of Recalls}_{i,j,t+n} = \beta_0 + \beta_1 \cdot \text{Percentage of Unionisation}_{i,t} + X_{i,j,t} + \text{Industry}_j \times \text{Year}_t + \varepsilon_{i,j,t}$$

if Number of Recalls > 0

where $X_{i,j,t}$ is the set of firm controls, $\text{Industry}_j \times \text{Year}_t$ refers to industry-year fixed effects, and $\varepsilon_{i,j,t}$ is the error term for firm i , industry j , and year t .

unionisation cases. Given the large number of possible specifications combining extensive and intensive margins of both recalls and unionization, I focus on a parsimonious set of models that separately examine (i) the effect of any union presence on the probability of any recall, and (ii) the effect of unionisation percentage on the number of recalls conditional on positive recalls, under alternative sets of controls and fixed effects.

A hypothesis presented by Kini et al. is that the product recalls may increase because higher cost of production would crowd out R&D expenditure and infrastructure improvements, lead to higher debt and/or impact firm productivity. If R&D intensity, book leverage and total factor productivity act as channels through which recalls increase, then adding them to the regression is bad control. However, if the recalls are not caused directly through this particular channel, yet these factors are correlated with both recalls and unionisation, excluding them would result in omitted variable bias. Therefore, I report the results with and without these controls.

I add a quadratic term for firm size, measured as log of market value of equity for the firm since its graphical relationship with cumulative recalls is non-linear and the coefficient on the quadratic term turns out to be statistically significant.

I cluster the standard errors at the firm level to account for conditional heteroskedasticity, which is a theoretical certainty in the linear probability model.

Results

From Table 1, any firm unionisation is associated with an approximate 9.7% points average increase in the likelihood of a recall in the next year, holding firm characteristics, and industry

- year fixed effects constant. The effect declines over time, with the probability of recall being approximately 6.7% points in three years. The estimates are statistically significant at the 1% level. The original paper predicted a higher magnitude of any unionisation on the frequency of recalls and the effect intensified over time. The estimate does not change by a large magnitude even when firm controls and industry fixed effects are removed. Hence, the concern regarding bad controls is alleviated.

In Table 2, the intensive margin effect of unionisation percentage on positive cumulative recalls fluctuates with and without industry fixed effects, but is significant at the 10 percent level with fixed effects. A percentage point increase in unionisation is associated with approx. 9.4 more cumulative recalls in the next year, conditional on it being a positive recalling firm, *ceteris paribus*. The estimate goes up to approximately 45 recalls in the three years of recalls, still significant at the 10% level. The magnitude of a similar estimate (not conditioning on positive recalls) in the original paper is much lower, but the significance is higher.

Table 1: Effect of any unionization on probability of any cumulative recalls

	(1) Prob(CumRecall t+1)	(2) t+1	(3) t+1	(4) t+3	(5) t+3	(6) t+3
Unionized (3 yrs)	0.0969*** (0.0151)	0.0984*** (0.0149)	0.0970*** (0.0159)	0.0676*** (0.0146)	0.0688*** (0.0144)	0.0658*** (0.0151)
Book leverage	0.00198* (0.00114)		-0.00172* (0.00101)	0.00512** (0.00210)		-0.00111 (0.00209)
Research and Development	-0.0479*** (0.00762)		0.00667 (0.00418)	-0.0597*** (0.0113)		0.00316 (0.00564)
Herfindahl-Hirschman Index	0.441** (0.209)	0.451** (0.192)	0.124*** (0.0288)	0.224 (0.168)	0.236 (0.162)	0.0935*** (0.0297)
Number of suppliers	0.00499*** (0.00145)	0.00504*** (0.00146)	0.00590*** (0.00154)	0.00413*** (0.00132)	0.00419*** (0.00133)	0.00519*** (0.00141)
Total Factor Productivity	0.00367* (0.00196)		0.00191 (0.00190)	0.00403* (0.00236)		0.000369 (0.00222)
Vertical Integration	-0.0234* (0.0127)	-0.0218* (0.0124)	-0.0727*** (0.0138)	-0.0386*** (0.0123)	-0.0367*** (0.0121)	-0.0873*** (0.0144)
Size	-0.0242*** (0.00399)	-0.0218*** (0.00360)	-0.0216*** (0.00413)	-0.0339*** (0.00541)	-0.0312*** (0.00494)	-0.0300*** (0.00562)
Size square	0.00429*** (0.000492)	0.00413*** (0.000463)	0.00386*** (0.000516)	0.00496*** (0.000612)	0.00477*** (0.000579)	0.00441*** (0.000642)
Year Fixed Effects	No	No	Yes	No	No	Yes
IndustryxYear Fixed Effects	Yes	Yes	No	Yes	Yes	No
Observations	32803	34270	32885	20566	21219	20669
R-squared	0.319	0.316	0.160	0.345	0.343	0.150
Adj. R-squared	0.295	0.293	0.160	0.316	0.315	0.150

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at firm level. The dependent variable for each column is a dummy variable which takes the value 1 if the firm has any cumulative recalls in the $t+n$ time period. Unionised is a dummy variable that takes the value 1 if the firm is unionised assuming a 3 year contract in the time period t .

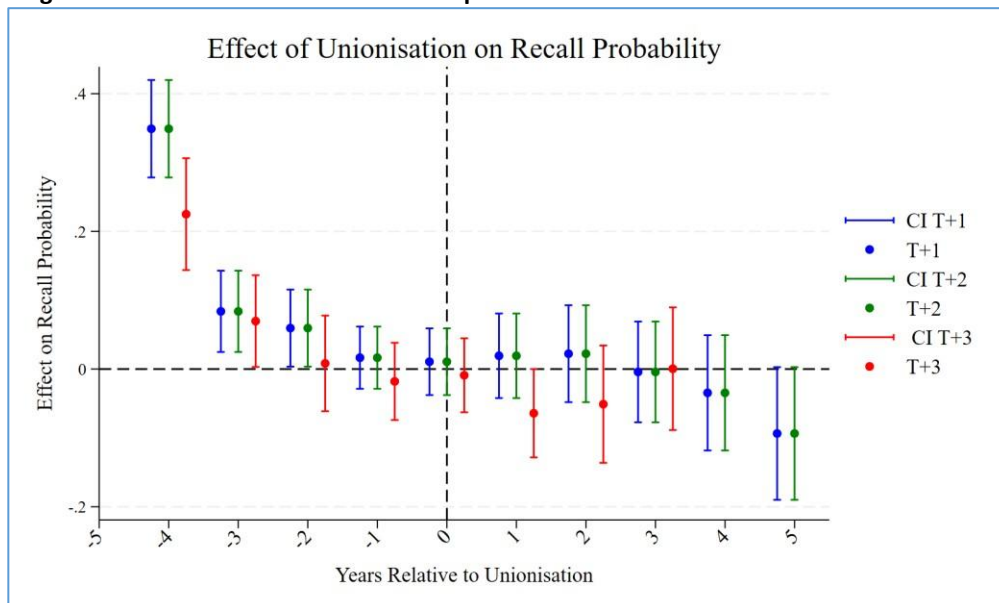
Table 2: Effect of Unionization percentage on number of positive cumulative recalls

	(1) CumRecalls(t+1)	(2) t+1	(3) t+1	(4) t+3	(5) t+3	(6) t+3
Pct Unionized (3 yrs)	9.276* (4.870)	9.581* (4.901)	5.868 (4.806)	43.96* (25.22)	46.22* (25.98)	27.10 (20.39)
Book leverage	0.768 (1.398)		2.800 (1.887)	2.847 (5.902)		11.67 (7.403)
Research and Development	-5.736 (4.211)		-0.798 (3.799)	-8.130 (20.80)		5.510 (18.38)
Herfindahl-Hirschman Index	13.93 (21.49)	13.64 (21.73)	-6.597*** (2.026)	132.4 (109.8)	131.9 (110.7)	-25.19*** (8.123)
Number of suppliers	0.0230 (0.0315)	0.0219 (0.0327)	0.0410 (0.0462)	0.0447 (0.0944)	0.0377 (0.103)	0.104 (0.145)
Total Factor Productivity	-0.674 (0.462)		-0.322 (0.441)	-4.149* (2.267)		-2.443 (1.917)
Vertical Integration	-1.361 (1.138)	-1.378 (1.128)	-1.851* (1.064)	-7.191 (5.115)	-7.155 (5.096)	-7.986* (4.151)
Size	-0.723 (1.122)	-0.760 (1.121)	-1.775* (0.997)	-1.454 (4.215)	-2.348 (4.259)	-5.058 (3.741)
Size square	0.114 (0.0819)	0.124 (0.0806)	0.164** (0.0711)	0.363 (0.301)	0.459 (0.302)	0.530** (0.261)
Year Fixed Effects	No	No	Yes	No	No	Yes
IndustryxYear Fixed Effects	Yes	Yes	No	Yes	Yes	No
Observations	1412	1412	1737	806	806	1011
R-squared	0.502	0.499	0.185	0.569	0.562	0.280
Adj. R-squared	0.383	0.380	0.176	0.465	0.459	0.268

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at firm level. The dependent variable for each column is the frequency of cumulative recalls, conditional on recalls > 0 in the $t+n$ time period. Percentage Unionised is the percentage of workers in the firm under union contracts assuming 3 year contracts, in the time period t .

Lastly, I plot estimates mapping out the probability of cumulative recalls over the years, taking the first instance of unionisation in a firm as a base. The plots use approximately 20003000 observations per estimate of $t + n$. Even after controlling for firm characteristics, industry-year fixed effects, firms which unionised had higher probability of recall 3 years prior to unionisation. The estimated probability of recall falls (insignificantly) after unionisation, in contrast to the Panel estimates. This casts doubts on the causality of the results in Table 1. I now turn to the RDD strategy to help understand these uncertain results.

Figure 1: Effect of unionisation on recall probabilities over time



RDD Specification

Empirical Strategy

In the election dataset, I use the 50% cut-off for unionisation to compare the impact of unionisation on recalls in firms which closely-won elections in contrast to recalls in firms which lost closely. I assume that there is no tampering with the election results, thus making the union win randomly assigned in a smaller bandwidth (15-20%) around the 50% cut-off. This randomisation ensures that any discontinuity in product recalls at the 50% cut-off is attributable to unionisation.

To check the validity of this assumption, I present a summary statistics table of the (preelection) sample characteristics to provide limited evidence of randomisation. In the 15% bandwidth around the cut-off, there is no statistically significant difference among firms in belonging to RTW versus non-RTW states and the agency of recall. There is no statistically significant likelihood for a firm to win an election in a particular year. In the 20% bandwidth, firms in RTW states were significantly more likely to suffer union election losses. I report my results with and without RTW status and firm-fixed effects for robustness.

Table 3: Pre-election firm characteristics across bandwidths

	35%–65%			30%–70%		
	Close Loss	Close Win	Test	Close Loss	Close Win	Test
N	274	202		351	258	
Year			0.325			0.125
2003	37 (13.5%)	23 (11.4%)		52 (14.8%)	35 (13.6%)	
2004	38 (13.9%)	19 (9.4%)		50 (14.2%)	22 (8.5%)	
2005	38 (13.9%)	34 (16.8%)		42 (12.0%)	44 (17.1%)	
2006	32 (11.7%)	15 (7.4%)		37 (10.5%)	21 (8.1%)	
2007	22 (8.0%)	19 (9.4%)		33 (9.4%)	22 (8.5%)	
2008	19 (6.9%)	20 (9.9%)		25 (7.1%)	22 (8.5%)	
2009	19 (6.9%)	11 (5.4%)		23 (6.6%)	13 (5.0%)	
2010	21 (7.7%)	12 (5.9%)		28 (8.0%)	15 (5.8%)	
2011	17 (6.2%)	22 (10.9%)		21 (6.0%)	28 (10.9%)	
2012	16 (5.8%)	16 (7.9%)		21 (6.0%)	20 (7.8%)	
2013	15 (5.5%)	11 (5.4%)		19 (5.4%)	16 (6.2%)	
Right-to-work			0.152			0.037
Non-RTW	200 (73.0%)	159 (78.7%)		256 (72.9%)	207 (80.2%)	
RTW	74 (27.0%)	43 (21.3%)		95 (27.1%)	51 (19.8%)	
FDA	0.620 (0.486)	0.558 (0.498)	0.198	0.617 (0.487)	0.598 (0.491)	0.651
NHTSA	0.090 (0.287)	0.094 (0.293)	0.895	0.102 (0.303)	0.085 (0.280)	0.516
CPSC	0.294 (0.457)	0.365 (0.483)	0.121	0.290 (0.455)	0.329 (0.471)	0.326

Right-to-work (RTW) is an indicator variable equal to 1 if the firm is headquartered in a U.S. state with right-to-work legislation and 0 otherwise (Non-RTW). FDA, NHTSA, and CPSC are indicator variables equal to 1 if the firm is subject to regulatory oversight from the U.S. Food and Drug Administration, the National Highway Traffic Safety Administration, or the Consumer Product Safety Commission, respectively, and 0 otherwise. Reported values are means with standard deviations in parentheses for continuous variables and counts with percentages in parentheses for categorical variables. P-values correspond to tests of differences in means between groups.

Since the cumulative number of recalls is zero for more than 80% observations and the positive number of recalls are right-skewed, I consider the same intensive and extensive margin approach as the panel specification. I report the estimates with and without firm and year dummies to isolate any possible within-firm effect and omit variations by year.

Extensive Margin:

$$\text{Recall dummy}_{i,j,t+n} = \beta_0 + \beta_1 \text{UnionWin}_{i,j,t} + \beta_2 \text{Dist}_{i,j,t} + \beta_3 (\text{UnionWin}_{i,j,t} \times \text{Dist}_{i,j,t}) + X_{i,j,t} + \text{Year}_t + \varepsilon_{i,j,t}$$

Intensive Margin:

$$\text{Frequency of positive recall}_{i,j,t+n} = \beta_0 + \beta_1 \text{UnionWin}_{i,j,t} + \beta_2 \text{Dist}_{i,j,t} + \beta_3 (\text{UnionWin}_{i,j,t} \times \text{Dist}_{i,j,t}) + X_{i,j,t} + \text{Year}_t + \varepsilon_{i,j,t}$$

where $X_{i,j,t}$ is the set of firm fixed effects, Year_t refers to year fixed effects, and $\varepsilon_{i,j,t}$ is the error term for firm i , industry j , and year t .

Results

Table 4: RDD Estimates: Effect of Unionization on probability of recall

	(1) 15% (t+1)	(2) 20% (t+1)	(3) 20% (t+1)	(4) 15% (t+3)	(5) 20% (t+3)	(6) 20% (t+3)
Union Win=1	-0.0138 (0.0733)	-0.0123 (0.0621)	0.0477 (0.0605)	-0.0776 (0.0831)	-0.0606 (0.0709)	0.0182 (0.0606)
election_dist	0.422 (0.502)	0.271 (0.333)	-0.00422 (0.339)	0.784 (0.586)	0.569 (0.392)	0.315 (0.360)
Union Win=1 × election_dist	-0.593 (0.822)	-0.314 (0.544)	-0.215 (0.501)	-0.980 (0.927)	-0.881 (0.608)	-0.965* (0.512)
Right-to-work		0.0370 (0.0357)			-0.00372 (0.0399)	
Constant	0.181*** (0.0447)	0.164*** (0.0390)	0.140*** (0.0387)	0.306*** (0.0526)	0.298*** (0.0468)	0.264*** (0.0403)
Firm Fixed Effects	No	No	Yes	No	No	Yes
Year Fixed Effects	No	No	Yes	No	No	Yes
Observations	476	609	609	476	609	609

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ Standard errors are robust. Union win is a dummy variable that takes 1 if the firm wins the union election in the time period t . election_dist is a variable defined as the percentage of votes for unionisation - 50, thus measuring distance from unionisation cut-off. The dependent variable in each column is a dummy variable that takes 1 if the firm has positive recalls in the $t+1$ (next) time period. The 15% bandwidth for instance includes firms which had 35-65% of votes for unionisation. The 20% bandwidth for instance includes firms which had 30-70% of votes for unionisation.

Table 4 indicates no statistically significant impact of unionisation on probability of product recalls. Table 5 suggests that unionisation results in 19.4 more recalls in the next three years for positive recalling firms, ceteris paribus, significant at the 10% level. However, the result is not significant with fixed effects. An important caveat is that the RDD is only valid for bandwidths close to the unionisation cut-off to measure the local average treatment effect (LATE) and the number of positive recalls are less than 20% of the sample. Even with a generous bandwidth of 20%, the number of positive recalls and total observations is quite low.

Table 5: RDD Estimates: Effect of Unionization on cumulative recalls

	(1) 15% (t+1)	(2) 20% (t+1)	(3) 20% (t+1)	(4) 15% (t+3)	(5) 20% (t+3)	(6) 20% (t+3)
Union Win=1	1.299 (3.347)	4.983 (3.788)	-0.340 (4.575)	11.55 (11.38)	19.43* (10.71)	4.173 (7.027)
election_dist	-1.971 (19.87)	4.784 (11.67)	33.36 (25.52)	-8.843 (25.76)	-38.42 (34.35)	72.53 (46.69)
Union Win=1 × election_dist	61.92 (53.48)	-15.03 (23.11)	-13.65 (39.03)	116.5 (147.3)	38.32 (88.41)	-74.65 (69.37)
Constant	3.695** (1.593)	4.028*** (1.474)	6.513** (3.136)	5.124*** (1.676)	3.658 (2.226)	15.62*** (3.221)
Firm Fixed Effects	No	No	Yes	No	No	Yes
Year Fixed Effects	No	No	Yes	No	No	Yes
Observations	72	92	92	111	138	138

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ Standard errors are robust. Union win is a dummy variable that equals 1 if the firm wins the union election in period t . election_dist is a variable defined as the percentage of votes for unionisation - 50, thus measuring distance from unionisation cut-off. The dependent variable in each column measures frequency of cumulative recalls, conditional on recalls > 0 in the $t+n$ time period.

Robustness

I check for reverse causality by including the lead and lag terms of the independent variable. I expect the main independent variable (unionisation) to be positive and statistically significant. The lag terms may be significant since it takes some time for unionisation to affect product quality. However, for a clean test of reverse causality, the leading terms must be statistically insignificant.

In the RDD, my next goal is to understand how other independent variables such as cost of production, operating leverage etc behave at the cut-off.

From Table 6, there seems to be mixed evidence related to reverse causality. One out of four leading terms is statistically significant at the 10% level. The relationship of probability and percentage of unionisation with probability of recall, is only significant with a gap of 2 time periods – not in the main unionisation variable which was significant in the Table 1. Perhaps most of the variation of the unionisation variables is actually captured by the lag and lead term. We noted before that both unionisation and recall variables have inflated zeros and fewer positive observations. If unionisation probability is persistent and hence highly correlated over time, the reverse-causality check model would suffer from multicollinearity. I find that indeed the probability and percentage of unionisation is highly correlated (upto 0.9) over time. Since multicollinearity may affect both the magnitude and significance of the estimates, we cannot conclusively rule for or against the presence of reverse causality in our panel specification.

Table 6: Effect of unionisation on recalls (leads and lags)

	(1) Freq Recall (t+1)	(2) Prob Recall (t+1)	(3) Freq Recall (t+1)	(4) Prob Recall (t+1)
Unionized (3 yrs) lag	0.109 (0.155)	0.0328** (0.0164)		
Unionized (3 yrs)	0.0564 (0.0971)	-0.0135 (0.0108)		
Unionized (3 yrs) lead	0.343 (0.221)	0.0391* (0.0212)		
Book leverage	-0.0102 (0.0199)	0.000988 (0.00165)	0.00446 (0.0233)	0.00299 (0.00198)
Research and Development	-0.314*** (0.103)	-0.0478*** (0.0109)	-0.368*** (0.115)	-0.0539*** (0.0120)
Herfindahl-Hirschman Index	-14.69 (12.13)	-2.566*** (0.824)	-15.71 (12.03)	-2.632*** (0.816)
Number of suppliers	0.0808* (0.0436)	0.00536*** (0.00152)	0.0787* (0.0433)	0.00525*** (0.00154)
Total Factor Productivity	-0.0368** (0.0183)	-0.00154 (0.00188)	-0.0392** (0.0192)	-0.00183 (0.00197)
Vertical Integration	-0.468*** (0.157)	-0.0324*** (0.0109)	-0.436*** (0.151)	-0.0298*** (0.0110)
Size	-0.384*** (0.102)	-0.0364*** (0.00631)	-0.421*** (0.110)	-0.0390*** (0.00681)
Size square	0.0459*** (0.0113)	0.00478*** (0.000698)	0.0500*** (0.0120)	0.00514*** (0.000741)
Pct Unionized (3 yrs) lead			-0.988 (0.909)	0.0283 (0.0463)
Pct Unionized (3 yrs)			0.767 (0.858)	-0.0374 (0.0375)
Pct Unionized (3 yrs) lag			1.519 (1.102)	0.0811* (0.0446)
Year Fixed Effects	No	No	No	No
IndustryxYear Fixed Effects	Yes	Yes	Yes	Yes
Observations	19342	19342	18734	18734
R-squared	0.264	0.347	0.268	0.345
Adj. R-squared	0.231	0.319	0.235	0.315

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at firm level. The dependent variable for columns 1 and 3 is the frequency of cumulative recalls in the $t+1$ time period. The dependent variable in columns 2 and 4 is the probability of positive recalls in the $t+1$ time period. Percentage Unionised is the percentage of workers in the firm under union contracts assuming 3 year contracts, in the time period t . Unionised is a dummy variable which takes 1 if the firm is unionised in the time period t , assuming 3 year contracts. The lag refers to the same variable in the previous time period, and the lead variable refers to the same variable in the next time period.

Table 7 reveals statistically significant discontinuities in the firms' outcome variables upon narrow union election wins. Unionised firms have higher costs of goods sold and operating leverages but no differential slope than firms with narrow election losses. Similarly, CSR and investments are lower, with no differential slope. These estimates could justify the original panel results but do not match with the RDD results. They suggest that unionisation results in

higher costs of goods to the firm, and operating leverage, crowding out investments and CSR, which align with the hypothesised factors that would deteriorate production quality.

Table 7: Impact of Unionisation on other outcome variables

	(1)	(2)	(3)	(4)
	COGS (t+1)	CSR (t+1)	Inv (t+1)	Op Lv (t+1)
Union win	0.0762*** (0.0286)	-0.404** (0.177)	-0.0319** (0.0130)	1.216* (0.669)
election_dist	-0.494** (0.209)	2.192 (1.407)	0.183* (0.0945)	-9.285* (5.263)
Union_win × election_dist	0.449 (0.321)	0.00387 (2.253)	-0.186 (0.145)	10.21 (8.198)
_cons	0.664*** (0.0191)	-0.0184 (0.106)	0.127*** (0.00870)	0.975** (0.438)
N	469	344	465	455

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are robust. COGS (t+1) denotes cost of goods sold in period $t+1$. CSR (t+1) denotes the firm's corporate social responsibility rating in period $t+1$. Inv (t+1) denotes ratio of capital expenditures plus R&D expenditures and other expenditures divided by book value of assets in period $t+1$. Op Lv (t+1) denotes operating leverage in period $t+1$.

Conclusion

This empirical study finds mixed evidence of the impact of unionisation on recalls. The DiD with fixed effects model indicates a positive impact of unionisation on the probability and frequency of recalls. However, in the year-wise plot, the parallel trends assumption might be violated and the coefficients are (insignificantly) negative. The reverse causality check on the DiD models neither confirms nor refutes the presence of a causal relationship.

The RDD does not find a significant relationship between unionisation and the probability or frequency of product recalls. However, the variables of the supposed causal channel-cost of goods, investment, operating leverage and CSR jump discontinuously in the expected directions. Given the significance of variables other than recalls in the RDD set-up, and the questionable validity of the panel results, I conclude that unionisation impacts the production costs of a firm but this may or may not translate into higher probability or frequency of recalls.

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